THE COMPOSITE OF DIFFERENTIAL OPERATORS OF ORDERS K AND K' HAS ORDER AT MOST K + K'

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The order of a differential operator. Definition used in the statement (lines 34903-34905) and in the induction at proof line 34914. Supplies the descending condition on the commutator with multiplication by g that the proof verifies.
- Additivity of the order under sums. Used at the last step of the counted proof (line 34913). Allows the displayed difference, once split into two composites of lower order, to be summed and bounded.

Lemma 0.1. *Let $R \to S$ be a ring map. Let L, M, N be S-modules. If $D : L \to M$ and $D' : M \to N$ are differential operators of order k and k', then $D' \circ D$ is a differential operator of order $k + k'$.*

Proof. Let $g \in S$. Then the map which sends $x \in L$ to

$$D'(D(gx)) - gD'(D(x)) = D'(D(gx)) - D'(gD(x)) + D'(gD(x)) - gD'(D(x))$$

is a sum of two compositions of differential operators of lower order. Hence the lemma follows by induction on $k + k'$. □

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